Ramanolla Shravani Satya Lakshmi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.26
Sri Chaitanya Jr.Kalasala Kondapur	Class XII (Telangana State board of Intermediate Education)	2023	98.70%
Sri Chaitanya Techno HS Balaji NGR, Medchal district	Class X (Board of Secondary Education, Telangana)	2021	10.00%

EXPERIENCE

Adobe | **Product Intern** | **Photoshop(Android)**

- Implemented a one-click "Try on my image" feature that automates complex, multi-step editing workflows using interactive before-and-after demonstration cards.
- Implemented a dynamic image analysis pipeline to evaluate user-uploaded images, automatically calibrating intermediate editing parameters to suit specific visual properties.
- Architected an interactive workspace that records manual user edits in real-time and seamlessly packages them into reusable transformation cards for scalable application to new images.

PROJECTS

Parallel Scheduling for Cloud Server Rental Systems | Research | Prof. Sathya Peri | C++

- Analyzed scheduling algorithms for **cloud server rentals with multi-dimensional resource** requirements.
- Developed approaches for non-clairvoyant scheduling models to handle unknown job durations.
- Evaluated resource utilization and applied parallelization techniques to improve execution scalability.

Intelligent Event Management & Engagement Platform | React.js, Node.js, MongoDB

- Developed a **cloud-based multi-tenant SaaS** platform to manage physical, virtual, and hybrid events.
- Implemented event lifecycles, ticketing, and payment workflows utilizing Role-Based Access Control.
- Integrated intelligent services such as chatbots and recommendation systems to enhance attendee engagement.

Inventory Mangement System | React.js, Express.js, PostgreSQL

- Developed a full-stack web application to manage inventory data, including staff, products, and warehouses across multiple inventory locations.
- Implemented Role-Based Access Control to restrict system access based on user role and assigned permissions.

RISC V Simulator | C++

- Designed a simulator to parse and execute RISC-V assembly instructions, featuring a real-time interface for memory and register states with integrated debugging controls.
- Developed a customizable cache simulator to analyze hit/miss statistics across variable sizes, associativities, and replacement policies.

ProbScript DSL Compiler | C++, Flex, Bison, CMake

- Developed the frontend of a custom probabilistic programming language as part of a Compilers course project.
- Implemented a Flex-based lexer for tokenizing and a Bison parser for syntax analysis and AST generation.
- Designed C++ class structures for AST representation and contributed to semantic validation.

Virtual Memory Management in GemOS | C

- Developed core memory management features that enable user programs to securely allocate and protect memory on demand.
- Implemented efficient memory-sharing mechanisms between processes to optimize system performance and reduce unnecessary data duplication.

Infinite Precision Arithmetic | C++, OOP

- Developed a C++ application to perform arithmetic opretions (addition, subtraction, multiplication and division) on integers and floating point numbers with infinite precision.

SKILLS

Programming Languages: C++, C, Python, SQL

Development/Tools: HTML, CSS, JavaScript, React.js, Express.js, Node.js, MySQL, PostgreSQL, MongoDB, Git, VS Code, GitHub, LaTeX, Lex, MakeFile, Numpy, Pandas, Pytorch, Matplotlib

RELEVANT COURSES

Computer Science : Algorithms, Data Structures, Database Management Systems, Operating Systems, Parallel & Concurrent Programming, Compilers, Computer Networks, Foundations of Machine Learning, Deep Learning, Computer Architecture, Software Engineering, Theory of Computation, Approximation Algorithms.

Mathematics : Probability & Random variables, Discrete Maths, Transform Techniques, Linear Algebra, Calculus, Statistics, Linear Optimization

POSITIONS OF RESPONSIBILITY

Teaching assistant for **CS3523:Operating Systems II** course, 2026

Deputy Coordinator for Sponsorship of Elan & Nvision, cultural and Tech fest of IIT Hyderabad.

EXTRACURRICULAR

Member of National Sports Organisation Athletics 2023 of IIT Hyderabad.